ensure all $G_M$ participate in training while maintaining efficiency, we randomly select one $G_M$ to calculate the SS_rate during each iteration of the model. The loss function is accordingly defined as follows:

$$\mathcal{L}_{\text{total}} = \underbrace{\mathcal{L}_{\text{task}}}_{\text{original loss}} + \alpha \cdot \left\| \text{SS_rate}(G_M^{(k)}) - \gamma \right\|_2^2 \tag{6}$$

where $G_M^{(k)}$ represents a randomly sampled feature network from multiple $G_M$ instances of model $M$, and $\gamma$ denotes the target value of self-similarity degree – specifically, the inherent self-similarity level of feature networks after standard training without regularization constraints.

## Experiments

To analyze the self-similarity levels of different models and explore the impact of self-similarity constraints on model performance, this paper designs experiments based on the following three aspects of the models in visual classification tasks:(1) Visualizing and analyzing the feature network ($G_M$) of different models;(2) Investigate the changes in SS_rate of different models before and after training, and use this as a reference to evaluate the impact of loss function constraints based on SS_rate on model training performance; (3) Explore the factors that influence model performance after applying SS_rate constraints.The source code are publicly available:
https://github.com/qcw-china/Self-similarity-Analysis-in-Deep-Neural-Networks

### Datasets and Models

**CIFAR-10**(Krizhevsky, Hinton et al. 2009), created by Alex Krizhevsky et al. in 2009, consists of 10 object categories (e.g., airplane, automobile, bird) with 6,000 32×32-pixel color images per category, totaling 60,000 images. The dataset is partitioned into 50,000 training samples and 10,000 test samples.

**CIFAR-100** extends CIFAR-10 with 100 fine-grained categories (e.g., "apple", "brown bear"), each containing 600 images (60,000 total). Maintaining identical 50,000/10,000 train-test partitioning, it introduces 20 superclasses (e.g., "mammals", "vehicles") to enable hierarchical classification tasks.

**Imagenette**(Howard et al. 2019), curated by Fast.ai in 2019, selects 10 easily distinguishable classes from ImageNet (e.g., "golf ball", "truck") comprising 13,000 images. Its distinguishing feature is higher resolution (default 160×160 pixels) providing richer visual granularity.

This paper adopts the module naming conventions from the timm library for all implemented models, as detailed in Table 1 which enumerates the specific architectural modules corresponding to each model type.

### $G_M$ Visualisation

We randomly sample 512 images from the ImageNet training set as model inputs to compute the $G_M$. Since the shallow features close to the input are highly sensitive to pre-training, their differences are significant and follow clear patterns. In contrast, the deep features near the output undergo relatively smaller changes after fine-tuning on the target task using the pre-trained model, making them less sensitive to the pre-training state. Therefore, Figure 2 presents the visualization results of $G_M$ from the shallow feature networks of different models.The visualization method used in this paper is Multidimensional Scaling (MDS) (Hout, Papesh, and Goldinger 2013), a multivariate statistical method that maps objects into a spatial representation based on their similarities or dissimilarities.

From Figure 2, it can be observed that the MLP architectures, such as Mixer and ResMLP, as well as the attention architecture ViT, exhibit strong low-dimensional manifold structures in their feature network $G_M$ before and after training, due to their global information aggregation mechanisms. This is specifically reflected in the significant clus-

| Model | Modules |
|---|---|
| ViT (small)(Dosovitskiy et al. 2020) | blocks.0–11 |
| ViT (base) | blocks.0–11 |
| PoolFormer(Yu et al. 2022) | stages.0–3 |
| PvT(Wang et al. 2021) | stages.0–3 |
| Resnet34 (He et al. 2016) | layer1–4 |
| Resnet50 | layer1–4 |
| Vgg(Simonyan and Zisserman 2014) | features.{1,4,9,14,19} |
| ResMLP(Touvron et al. 2022) | blocks.0–11 |
| MLP-Mixer(Tolstikhin et al. 2021) | blocks.0–11 |

Table 1: Model Architectures and Corresponding Modules

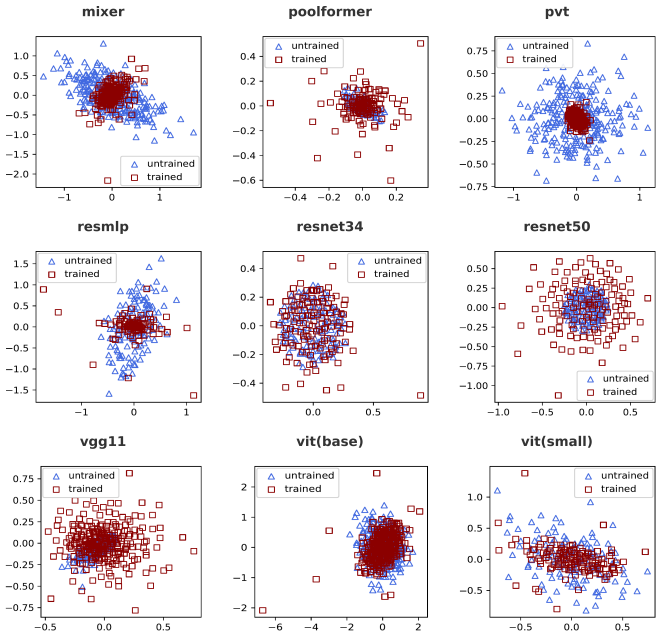


Figure 2: Comparative Visualization of $G_M$ Distributions Across Multiple Architectures Pre- and Post-Training

tering and directionality of the features.

In contrast, convolutional architectures such as ResNet and Vgg, along with PVT, which have local receptive fields limiting information interaction, show feature network $G_M$ that are more uniformly distributed. This is manifested as isotropic diffusion of features with no dominant direction. The $G_M$ of the PoolFormer model exhibit a strong low-dimensional manifold structure only when the model is untrained, but show a more uniform distribution after pre-training. This is because PoolFormer employs local static mixing operations combined with nonlinear transformations, which disrupt the model's initial global operation mechanism. It can be seen that their core difference lies in the information processing mechanism: architectures with local inductive bias lead to more uniform $G_M$ distributions, whereas architectures with global information aggregation lead to more linear distribution patterns in $G_M$.

## Quantitative Analysis of Self-Similarity in $\mathbf{G_M}$

This section delves into the self-similarity of $G_M$. To this end, this study proposes a quantitative metric for self-similarity—the Self-Similarity Rate (SS_rate). Table 2 presents the average SS_rate values of $G_M$ across all hidden layers for each model in both the untrained and pre-trained states.

| Model | Pre-train Avg | Post-train Avg |
|---|---|---|
| **ViT (small)** | 0.13 | **0.66** |
| **ViT (base)** | 0.13 | **0.46** |
| **PoolFormer** | 0.34 | **0.43** |
| **PvT** | 0.17 | **0.48** |
| **Resnet34** | 0.22 | 0.17 |
| **Resnet50** | 0.22 | **0.33** |
| **Vgg** | 0.19 | 0.18 |
| **ResMLP** | 0.09 | **0.40** |
| **MLP-Mixer** | 0.13 | **0.66** |

Table 2: Model-wise SS_rate Averages before and after Training

Table 2 shows the changes in SS_rate for all models before and after training. In the majority of models, the SS_rate increases after training. This indicates that as training progresses, the models gradually transition from a state of high self-similarity to one of lower self-similarity, which aligns with the core optimization process of neural networks evolving from an initial random state toward efficient feature representations.

Since SS_rate encodes the relationships between neuron outputs, the high self-similarity observed in the early stages of training reflects a convergence in neurons' activation patterns toward input patterns such as edges and textures. This results in high redundancy in the hidden feature maps across spatial or channel dimensions. As training proceeds, the weights of neurons corresponding to important features are significantly enhanced, while redundant neurons are updated to a lesser extent. This breaks the initial symmetry, reduces the redundancy in hidden feature maps, and consequently lowers the degree of self-similarity.

Next, we design experiments to verify whether constraining SS_rate during training—specifically by promoting its convergence toward post-training target values—positively impacts model accuracy at convergence. In the experiments, we set the value of the hyperparameter $\alpha$ to the order of magnitude of $10^{-4}$, and each set of experiments is conducted ten times to compute the mean and standard deviation. Our constraint implementation employs a straightforward mean squared error (MSE) loss penalty, leveraging the differentiability and computational efficiency of SS_rate that we previously emphasized.

As demonstrated in Table 3, the regularization constraint yields significant performance improvements exclusively for ResMLP, MLP-Mixer, and ViT-series models (averaging 1.2-5.3% gain), while exhibiting negligible impact on convolution-dominated architectures like Vgg/ResNet. This phenomenon contrasts sharply with the universal SS_rate elevation trend observed in Table 1, revealing the model-structure-dependent efficacy of self-similarity constraints.

## Architectural Self-Similarity: The Decisive Factor in Constraint Efficacy

Subsequently, we design experiments to validate the correspondence between constrained optimization outcomes and two fundamental properties across different hidden layers: (1) statistical scale invariance (Mahoney and Martin 2019)Table 4 and (2) geometric invariance (Grassberger and Procaccia 1983)(Theiler 1990) Table 5. Statistical Scale Invariance is quantified by computing feature sparsity and fitting power-law distributions per layer, with invariance strength measured by the standard deviation of power-law exponents across layers. Geometric Invariance is evaluated using the correlation dimension algorithm, where invariance degree is determined by relative fluctuation of fractal dimensions derived from log-log linear fits of correlation integrals.

This investigation aims to establish that intrinsic architectural self-similarity directly governs the effectiveness of our self-similarity constraints. Methodologies for measuring these invariance properties are detailed in the Appendix B,C.

Table 3 shows that the models exhibiting significant performance improvements under the $S^2$-Con regularization constraint—ResMLP, MLP-Mixer, and ViT—are precisely those that demonstrate better statistical scale invariance and geometric invariance (Table 4, 5). Statistical scale invariance and geometric invariance are key indicators for evaluating whether a network possesses self-similarity. Therefore, the experimental results indicate that for models that inherently exhibit good self-similarity, imposing such constraints can further enhance their existing performance advantages.

Moreover, the SS_rate constraint is designed to strengthen self-similarity at the global scale. However, the PoolFormer model, which relies on local features, cannot effectively respond to such global constraints (as analyzed in Figure 2). The Vgg model, which inherently has poor self-similarity, experiences a slight performance drop under the constraint, demonstrating that forcibly altering the local information interaction patterns in models with poor self-similarity can

| | Model | CIFAR10 | | CIFAR100 | | Imagenette | |
|---|---|---|---|---|---|---|---|
| | | baseline | $S^2$-Con | baseline | $S^2$-Con | baseline | $S^2$-Con |
| MLP | ResMLP↑ | 87.35 (±0.34)<br>0.29 (±0.00) | **88.17 (±0.09)**<br>0.36 (±0.00) | 62.16 (±0.11)<br>0.24 (±0.00) | **64.42 (±0.16)**<br>0.30 (±0.00) | 84.63 (±0.16)<br>0.21 (±0.01) | **85.89 (±0.18)**<br>0.37 (±0.01) |
| | MLP-Mixer↑ | 86.81 (±0.13)<br>0.15 (±0.00) | **88.27 (±0.13)**<br>0.25 (±0.00) | 61.50 (±0.17)<br>0.31 (±0.00) | **62.53 (±0.19)**<br>0.29 (±0.00) | 82.96 (±0.14)<br>0.29 (±0.00) | 82.16 (±0.19)<br>0.19 (±0.00) |
| CNN | Vgg | 89.98 (±0.07)<br>0.29 (±0.01) | 89.83 (±0.09)<br>0.33 (±0.01) | 60.48 (±0.18)<br>0.19 (±0.01) | 60.06 (±0.12)<br>0.17 (±0.01) | 90.38 (±0.17)<br>0.30 (±0.01) | 90.23 (±0.16)<br>0.29 (±0.01) |
| | Resnet34 | 93.90 (±0.07)<br>0.25 (±0.01) | **94.00** (±0.10)<br>0.24 (±0.00) | 74.23 (±0.26)<br>0.24 (±0.00) | **74.35** (±0.31)<br>0.24 (±0.00) | 89.20 (±0.23)<br>0.19 (±0.01) | **89.25** (±0.14)<br>0.20 (±0.01) |
| | Resnet50 | 94.11 (±0.09)<br>0.22 (±0.00) | **94.13** (±0.13)<br>0.22 (±0.00) | 73.77 (±0.29)<br>0.24 (±0.00) | **73.87** (±0.22)<br>0.23 (±0.00) | 90.50 (±0.16)<br>0.19 (±0.01) | 90.34 (±0.24)<br>0.17 (±0.00) |
| Attention | ViT (small)↑ | 89.99 (±0.14)<br>0.49 (±0.00) | **90.50 (±0.19)**<br>0.51 (±0.00) | 65.75 (±0.16)<br>0.60 (±0.00) | **67.69 (±0.18)**<br>0.52 (±0.00) | 81.71 (±0.33)<br>0.49 (±0.00) | **83.74 (±0.28)**<br>0.48 (±0.01) |
| | ViT (base)↑ | 90.29 (±0.44)<br>0.55 (±0.00) | **93.21 (±0.23)**<br>0.58 (±0.00) | 61.40 (±0.22)<br>0.41 (±0.01) | **67.15 (±0.22)**<br>0.45 (±0.00) | 77.88 (±0.40)<br>0.29 (±0.00) | **82.93 (±0.24)**<br>0.39 (±0.01) |
| | PoolFormer | 92.85 (±0.07)<br>0.43 (±0.00) | 91.50 (±0.16)<br>0.55 (±0.00) | 71.32 (±0.17)<br>0.46 (±0.00) | 68.56 (±0.50)<br>0.52 (±0.00) | 82.76 (±0.23)<br>0.64 (±0.00) | **84.70 (±0.29)**<br>0.48 (±0.01) |
| | PvT | 94.55 (±0.09)<br>0.36 (±0.01) | 94.43 (±0.07)<br>0.47 (±0.01) | 74.73 (±0.09)<br>0.44 (±0.01) | 74.45 (±0.18)<br>0.45 (±0.00) | 89.48 (±0.15)<br>0.31 (±0.01) | **89.64 (±0.17)**<br>0.35 (±0.01) |

Table 3: Model performance comparisons across multiple datasets under baseline settings versus SS_rate-constrained optimization (denoted as $S^2$-Con).

| Model | CIFAR10 | CIFAR100 | Imagenette | Avg |
|---|---|---|---|---|
| **MLP-Mixer** | 0.0758 | 0.0786 | 0.2489 | 0.1344 |
| **ViT** | 0.1521 | 0.1779 | 0.1722 | 0.1674 |
| **PoolFormer** | 0.1915 | 0.1886 | 0.3771 | 0.2524 |
| **Resnet** | 0.2688 | 0.1683 | 0.5158 | 0.3176 |
| **ResMLP** | 0.3453 | 0.3404 | 0.3404 | 0.3420 |
| **PvT** | 0.4909 | 0.4506 | 0.7503 | 0.5639 |
| **Vgg** | 0.7898 | 0.6066 | 0.7482 | 0.7149 |

Table 4: Statistical Scale Invariance Rankings

| Model | CIFAR10 | CIFAR100 | Imagenette | Avg |
|---|---|---|---|---|
| **MLP-Mixer** | 0.2432 | 0.1390 | 0.2270 | 0.2031 |
| **ResMLP** | 0.2608 | 0.2137 | 0.3417 | 0.2721 |
| **PoolFormer** | 0.2868 | 0.2126 | 0.3212 | 0.2735 |
| **ViT** | 0.2456 | 0.2192 | 0.3669 | 0.2772 |
| **Vgg** | 0.4434 | 0.2422 | 0.2655 | 0.3170 |
| **Resnet** | 0.3093 | 0.4126 | 0.6399 | 0.4540 |
| **PvT** | 0.4163 | 0.5249 | 0.6216 | 0.5210 |

Table 5: Geometric Invariance Rankings

disrupt their optimization process.

## Conclusion

This work proposes $G_M$, a topological feature network constructed from hidden-layer activations – as a new paradigm for complex network representation of neural networks. Transcending conventional structural analysis paradigms,$G_M$ achieves unified modeling across diverse architectures (MLP/CNN/Attention) by capturing dynamic relationships within neuron activation patterns. Experimental results demonstrate that $G_M$ effectively characterizes the dynamics of model parameter distributions. It is observed that the limited information exchange inherent to local receptive fields drives the $G_M$ toward a uniform distribution, while architectures employing global information aggregation exhibit prominent low-dimensional manifold structures. Throughout the training process, the self-similarity of $G_M$ generally diminishes. An self-similarity constraint based on the SS_rate metric achieved significant performance improvements in models with high self-similarity.

While $G_M$ provides a novel tool for investigating neural network topological properties, the dynamic evolution of self-similarity requires more refined modeling. Current constraint strategies employing fixed self-similarity targets may constrain the autonomous optimization pathways of models. Future research could design time-variant objective functions to dynamically adapt to requirements across distinct training phases, and explore applications of $G_M$ in studying topological characteristics like small-world properties.

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